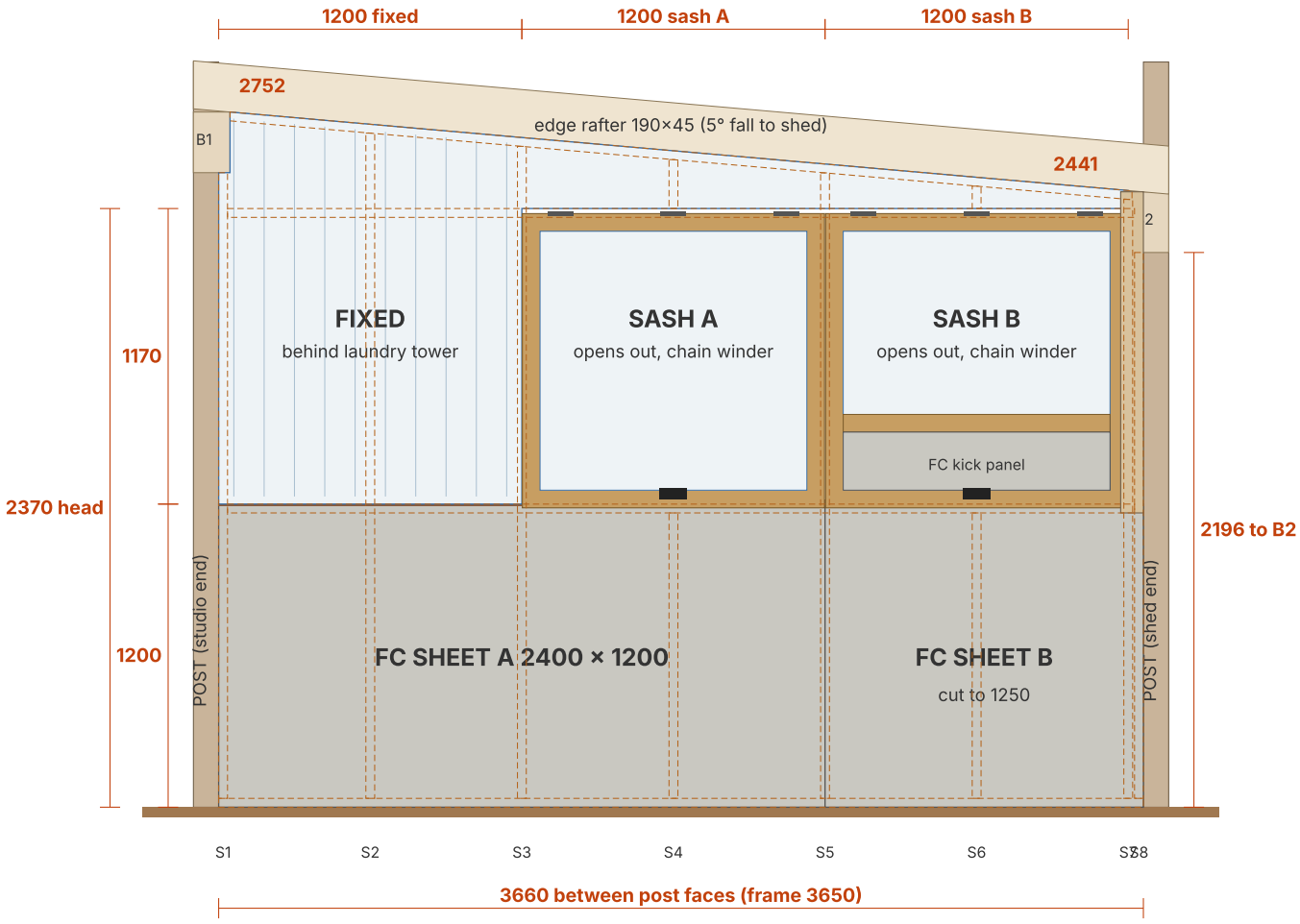
Fence-end wall and windows: build plans

87 Prince St, Mullumbimby · between the two fence-end posts · FC on the deck side to 1 200 · one fixed twinwall bay, two opening twinwall sashes on chain winders · all mm · P28, 6 Nov 2026

ELEVATION, SEEN FROM THE DECK



6 mm FC, clear sealed (deck face) · Opal twinwall (fence face) · Sash frame · 70×35 H3 frame · Sill board

SET-OUT

- **Studs** at 600 centres from the studio-end post: S1 against the post, S2 600, S3 1 200, S4 1 800, S5 2 400, S6 3 000, S7 3 600, S8 against the shed-end post.
- **S4 and S6** (the middle of each sash bay) stop at the sill nogging and restart above the head: clear openings 1 165 wide, room for the winders.
- **Two nogging rows, on flat, full 70 depth:** sill 1 165–1 200 and head 2 335–2 370. Run them continuous (1 165) across the two sash bays.
- **FC** on the deck face, deck to 1 200: sheet A full from the studio end, sheet B cut to 1 250, joint on S5.

UPPER WALL

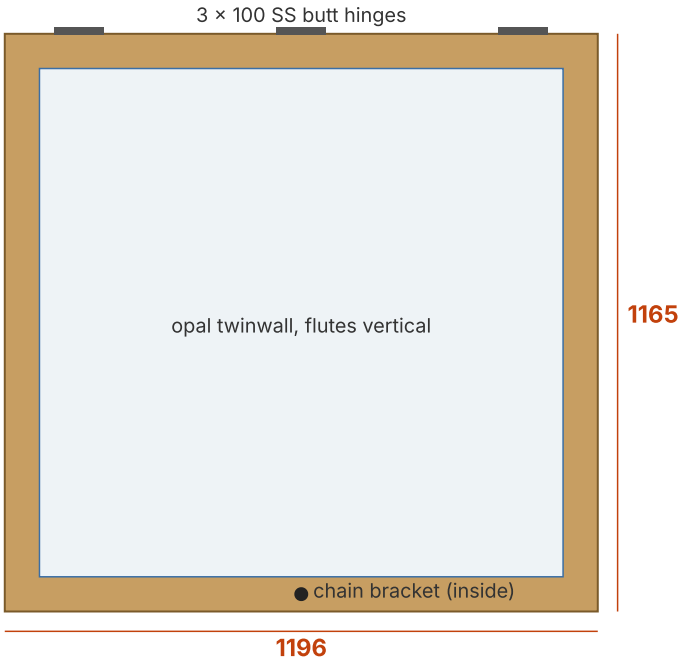
- **Bay 1 (0–1 200), fixed:** it's behind the 2.05 m laundry tower, so it never needs to open. Twinwall screwed to the fence face of the frame, 1 200 up to the rafter.
- **Bays 2 and 3: sashes A and B,** top-hung on the fence face, opening outward (fence 1.2 m clear) on lockable chain winders. Sash B, behind the wok, has an FC kick panel.
- **Above the head:** a fixed raked twinwall strip up to the rafter, with a head flashing over the hinges.
- **Shed end:** the last 60 is covered by a 90×19 cover batten.

Text Ollie before building. A solid wall catches more wind than the batten screen he checked. Ask him about the wall and how it ties to the posts, beams and rafter.

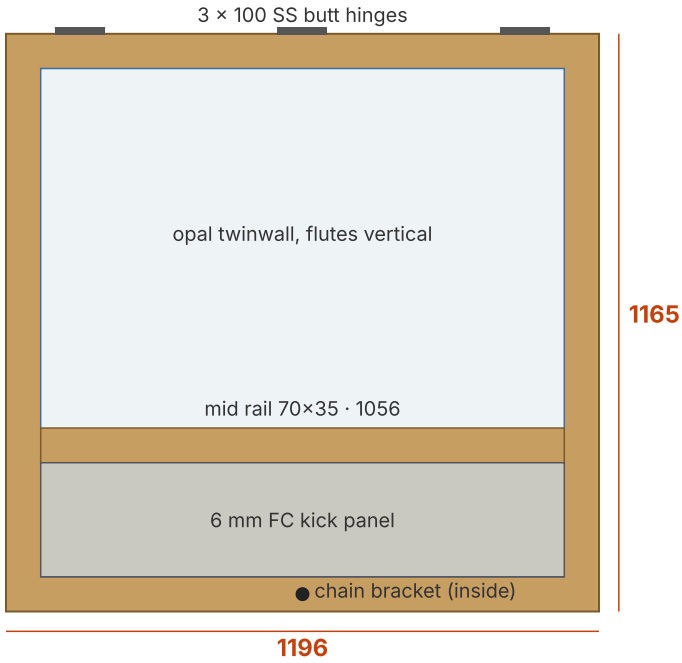
Windows: sash and section details

Two sashes, each 1 196 wide × 1 165 tall, seen from the fence side. Built flat on a bench, then hung.

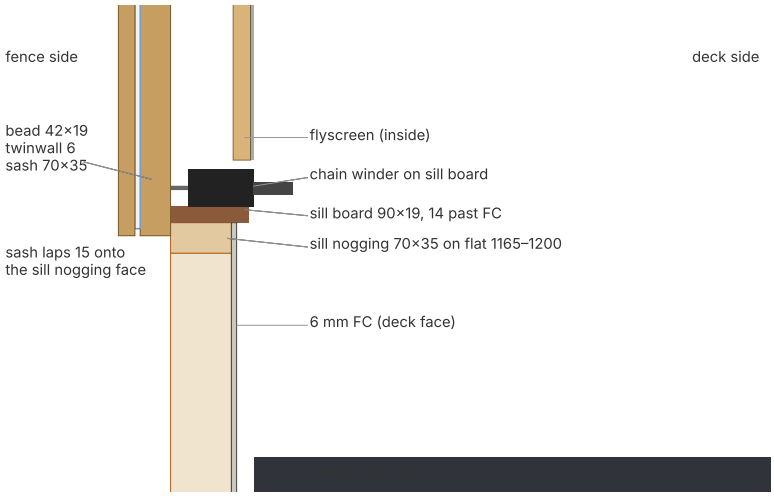
SASH A (BAY 2)



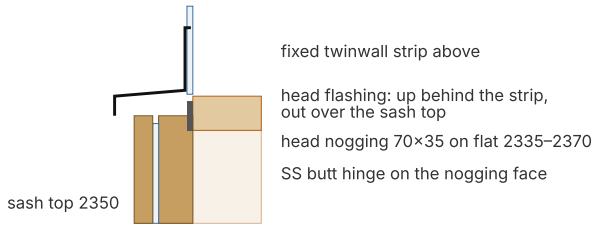
SASH B (BAY 3, BEHIND THE WOK)



SECTION THROUGH THE SILL



SECTION THROUGH THE HEAD



HOW THE WINDOW WORKS

- Lean over the bench and wind the handle on the sill board: 19 turns opens it to 300 at the bottom. A clip limits it to 100 or 150 on windy days.
- Wound shut, it pulls the sash tight onto the seal and locks it (keyed). No bolts or stays needed.
- The flyscreen sits above the winder, so you can wind with the screen in.
- Sash B's winder is behind the wok stove: wind it open before you light up.

SASH BUILD

1. Frame: 70×35 H3 on flat. Two stiles at 1 165, two rails at 1 056 between them (sash B also gets a 1 056 mid rail with its top 370 up from the bottom). Glue every joint and drive two 100 mm batten screws per corner through the stiles into the rail ends. Pre-drill, and check the diagonals.
2. Seal all timber, including end grain. Tape the twinwall: solid aluminium tape on the top edge, breather tape on the bottom edge.
3. Lay the twinwall on the outside face (sash A: 1 180 × 1 149; sash B: 1 180 × 822 above the mid rail, and a 1 180 × 327 FC kick panel below it). Clamp it with 42×19 H3 beads round the edge (and over the mid rail on B), screwed every 300 through 3 mm oversize holes in the poly.
4. Hang each sash on three 100 mm stainless butt hinges on the outside face of the head nogging, lapping 15 over the studs, sill and head. Stick foam weatherseal on the frame faces where the sash closes.
5. Fit the winder centred on the sill board, and the chain bracket centred on the inside face of the bottom rail. Adjust it so it winds in tight.

Materials, cut lists and build order

Buy 1 200-wide twinwall: 1 220-wide sheets won't line up with the studs. 70×35 and 42×19 are common at Old Mill. The chain winders are a Bunnings special order.

MATERIAL LIST

Item	Qty
70×35 H3 (MGP10 or F7) × 2.7 m: studs	5
70×35 H3 × 2.4 m: studs S7, S8	2
70×35 H3 × 3.6 m: sill and head noggings	2
70×35 H3 × 4.2 m: bottom and top plates	2
70×35 H3 × 6.0 m + 5.4 m: sash frames	1 + 1
42×19 H3 DAR × 5.4 m + 1.8 m: sash beads	2 + 1
90×19 H3 DAR (or merbau) × 5.4 m: sill board + cover batten	1
6 mm fibre cement 2 400 × 1 200	2
Opal twinwall polycarbonate 1 200 × 2 400	2
Polycarb U-cap (fixed panel edges), 2.1 m	3
Solid aluminium tape + breather tape	1 each
Head flashing 2.4 m (Zincalume or Colorbond)	1
Whitco CYL4 lockable chain winder (Bunnings special order, $73.21)	2
100 mm stainless butt hinges	6
Foam or EPDM weatherseal, 6–9 mm	≈10 m
Poly screws with EPDM washers, 12g × 45	≈60
Galv FC nails 40 × 2.8	500 g
14g × 100 batten screws (plates, posts, sash corners)	≈50
75 galv nails or timber screws (frame)	≈60
Exterior PU glue, clear FC sealer 1–2 L, end-seal, timber sealer	1 each
Flyscreens: 45×32 hoop pine ≈9 m (ripped to 20 + 10), fibreglass mesh ≈2.4 m of 1.2 m roll, 8 turn buttons	2 screens

TIMBER CUT LIST

Piece	Length	From
Bottom plate	3 650	4.2 m
Top plate (raked, under rafter)	≈3 585	4.2 m
S1 (to B1, square top)	2 467	2.7 m
S2 · S3 · S5 (5° top)	2 624 · 2 571 · 2 466	2.7 m each
S4 + S6 lower stubs, S4 + S6 upper stubs	1 130 × 2, 184, 79	one 2.7 m
S7 (5° top) · S8 (to B2)	2 361 · 2 151	2.4 m each
Sill nogging row (on flat)	548, 565, 1 165, 1 165	3.6 m
Head nogging row (on flat)	548, 565, 1 165, 1 165	3.6 m
Sash A frame	1 165 × 2, 1 056 × 2	5.4 m
Sash B frame + mid rail	1 165 × 2, 1 056 × 3	6.0 m
Beads, each sash	1 165 × 2, 1 112 × 2	5.4 m each
Sash B mid bead	1 112	1.8 m
Sill board · cover batten	3 660 · ≈1 280	5.4 m

Provisional lengths, taken from the model, with the whole frame built 10 under the rafter and 5 off each post. Finalise them from the site measurements.

SHEET CUT LIST

Sheet	Cut
FC A	Full 2 400 × 1 200
FC B	1 250 × 1 200; from the 1 150 × 1 200 offcut cut sash B's kick panel 1 180 × 327
Twinwall 1	Fixed bay: 1 200 wide, 1 552 tall at the studio end falling to 1 451, with a 45 × 240 notch for B1. From the offcut: the raked strip over the sashes, 2 pieces 1 200 wide (281→176 and 176→71)
Twinwall 2	Sash A panel 1 180 × 1 149, then sash B panel 1 180 × 822

BUILD ORDER (PREFAB AND TILT-UP)

1. Measure on site: rafter heights over the wall line every 600, the width between the posts at three heights, the B1 and B2 ply depths, and any bolt heads. Finalise the cut list with 10 clearance at the top and 5 at each post.
2. Build the whole frame flat on the deck, top towards the deck: bottom plate 3 650, the studs and stubs, the raked top plate with its ends notched under the B1 and B2 plies (and round any nuts), and the sill and head nogging rows on flat. Square it by the diagonals. End-seal every cut.
3. Nail the FC on face-up while it's flat: 6 above the plate underside, nails at 200 on the edges and 300 in the field, 12 in from the edges. Seal it.
4. Stand it up between the posts (3 people). Tap folding wedges in over each stud until it's tight to the rafter, then plumb it. Screw the end studs into the posts (pre-drill), up through the top plate and wedges into the rafter every 600, and the bottom plate into the edge joist every 450, all 14g × 100.
5. Fix the sill board flush with the fence face and 14 proud of the FC inside. Seal it.
6. From the fence side, fix the bay 1 twinwall and the raked strip over the head. U-cap the edges and screw through oversize holes, snug only. Fit the cover batten over S7 and S8.
7. Build and hang the sashes (previous page). Fit the head flashing up behind the raked strip and over the sash tops.
8. Fit the winders and seals, test the windows, then fit the flyscreens inside with turn buttons.

Heat: sash B's FC kick panel keeps the poly about 420 above the wok rings. Still check the stove manual for clearance to combustibles, and open sash B while you cook.